

Problem Statement

Evaluate the following limit from the image "dailyintegral-limit-limit-1x1-2026-07-21.jpg":

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{1}{4k+1} + \frac{1}{4k+2} - \frac{1}{4k+3} - \frac{1}{4k+4} \right)$$

Detailed Solution

This limit represents the sum of an infinite series whose terms alternate signs in pairs. We can evaluate it elegantly using two distinct approaches: using integral representations of individual terms and using the Taylor series expansion of $\ln(1+x)$.

Step 1: Express terms using integral representations

Recall the standard integral identity for fractions of the form $\frac{1}{m}$:

$$\frac{1}{m} = \int_0^1 x^{m-1} dx \quad \text{for } m > 0$$

Applying this identity to each term inside the summation gives:

$$\begin{aligned} \frac{1}{4k+1} &= \int_0^1 x^{4k} dx \\ \frac{1}{4k+2} &= \int_0^1 x^{4k+1} dx \\ \frac{1}{4k+3} &= \int_0^1 x^{4k+2} dx \\ \frac{1}{4k+4} &= \int_0^1 x^{4k+3} dx \end{aligned}$$

Step 2: Combine the terms inside the sum and integral

Substituting these integrals back into the finite sum yields:

$$S_n = \sum_{k=1}^n \int_0^1 (x^{4k} + x^{4k+1} - x^{4k+2} - x^{4k+3}) dx$$

Factoring out x^{4k} from the integrand:

$$S_n = \int_0^1 (1 + x - x^2 - x^3) \sum_{k=1}^n x^{4k} dx$$

Notice that $1 + x - x^2 - x^3 = (1 + x)(1 - x^2) = (1 - x)(1 + x)^2$. Furthermore, the inner sum is a finite geometric series with initial term x^4 and common ratio x^4 :

$$\sum_{k=1}^n x^{4k} = x^4 \frac{1 - x^{4n}}{1 - x^4}$$

Step 3: Evaluate the infinite limit under the integral

Taking the limit as $n \rightarrow \infty$, since $|x| < 1$ almost everywhere on the interval $(0, 1)$, we have $x^{4n} \rightarrow 0$. By the Monotone Convergence Theorem, we can interchange the limit and integral:

$$S = \int_0^1 (1 - x)(1 + x)^2 \cdot \frac{x^4}{1 - x^4} dx$$

Since $1 - x^4 = (1 - x)(1 + x)(1 + x^2)$, we simplify the integrand:

$$S = \int_0^1 \frac{(1 - x)(1 + x)^2 x^4}{(1 - x)(1 + x)(1 + x^2)} dx = \int_0^1 \frac{(1 + x)x^4}{1 + x^2} dx$$

Step 4: Perform polynomial division and integrate

Expanding the numerator gives:

$$S = \int_0^1 \frac{x^5 + x^4}{1 + x^2} dx$$

Performing polynomial division on the integrand:

$$\frac{x^5 + x^4}{x^2 + 1} = x^3 + x^2 - x - 1 + \frac{x + 1}{x^2 + 1}$$

Now, integrate each term individually over $[0, 1]$:

$$S = \int_0^1 \left(x^3 + x^2 - x - 1 + \frac{x}{x^2 + 1} + \frac{1}{x^2 + 1} \right) dx$$

Evaluating the antiderivative at the limits 0 and 1:

$$S = \left[\frac{x^4}{4} + \frac{x^3}{3} - \frac{x^2}{2} - x + \frac{1}{2} \ln(1 + x^2) + \arctan(x) \right]_0^1$$

$$S = \left(\frac{1}{4} + \frac{1}{3} - \frac{1}{2} - 1 + \frac{1}{2} \ln(2) + \frac{\pi}{4} \right) - (0)$$

Combining the rational constants:

$$\frac{1}{4} + \frac{1}{3} - \frac{1}{2} - 1 = \frac{3 + 4 - 6 - 12}{12} = -\frac{11}{12}$$

Thus, the sum evaluates to:

$$S = \frac{\pi}{4} + \frac{1}{2} \ln(2) - \frac{11}{12}$$

Alternative Method: Relation to the Mercator Series for $\ln(2)$

We can also express the infinite series directly by manipulating the alternating harmonic series:

$$\sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln(2)$$

Our given series S corresponds to the terms starting from $k = 1$, which misses the initial four terms for $k = 0$:

$$S = \sum_{m=5}^{\infty} c_m, \quad \text{where } c_m = \begin{cases} 1/m & m \equiv 1, 2 \pmod{4} \\ -1/m & m \equiv 3, 0 \pmod{4} \end{cases}$$

Adding and subtracting the missing terms $1 + \frac{1}{2} - \frac{1}{3} - \frac{1}{4} = \frac{11}{12}$ alongside the arctan series expansion $\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}$ similarly reduces to the exact identity:

$$S = \frac{\pi}{4} + \frac{1}{2} \ln(2) - \frac{11}{12}$$

Final Answer:

$$\frac{\pi}{4} + \frac{1}{2} \ln(2) - \frac{11}{12}$$